

MM-SAG: Multi-Modal Semantic-Aware Graph for Multi-Label Image Classification of Kitchen Waste

Hai Qin, Jintao Li, Qiaokang Liang, *Senior Member, IEEE*, Wei Sun, Dan Zhang, *Senior Member, IEEE*, Yaonan Wang

Abstract—Multi-label image classification aims to recognize multiple object labels within an image. In the field of intelligent waste sorting, efficient classification can enhance the accuracy of robotic sorting. However, most existing waste classification tasks are single-label, and there is limited research on multi-label classification of urban kitchen waste, especially in complex backgrounds and diverse categories. To address this, we propose a Multi-Modal Semantic-Aware Graph (MM-SAG) framework, which includes a semantic-aware module designed for instance-level label relationship mining. The captured semantic features are then processed through graph convolution to generate a label correlation matrix, enhancing the efficiency and effectiveness of label correlation mining. To improve the integration of visual and linguistic modalities, we design an improved multi-head attention mechanism module. This module re-encodes and aligns visual and textual features, further enhancing feature extraction capabilities. Experimental results show that our proposed method achieves a mean Average Precision (mAP) of 83.1% on the MLKW dataset, delivering state-of-the-art performance. The method's strong generalization capability is also validated on public datasets VOC2007 and MS-COCO. The source code and dataset are available at our GitHub repository: <https://github.com/yulle13/MMSAG.git>.

Index Terms—Urban kitchen waste; Multi-label classification; Multi-modal; Multi-head attention.

I. INTRODUCTION

SOCIAL progress is reflected not only in the development of advanced technologies but also in the improvement of human health and harmonious development with nature. Environmental sanitation in residential areas is a crucial factor. With the improvement of residents' quality of life, the amount of kitchen waste generated in daily activities has surged [1], [2], [3]. As environmental awareness deepens, waste sorting and recycling have become new trends. Currently, kitchen

waste treatment methods have evolved through stages such as composting or use as animal feed [4], landfill disposal with other wastes, and secondary development and utilization. The microorganisms and water content in kitchen waste ensure its good anaerobic digestion, producing biogas [5]. However, due to a lack of strong sorting awareness, urban kitchen waste often contains other non-degradable objects, including metal products, cans, cartons, bottles, and other types of waste. If non-degradable materials are not removed from kitchen waste, it can severely affect subsequent biochemical treatment processes. Therefore, this paper focuses on multi-label image classification of kitchen waste, making the first attempt to study kitchen waste multi-label classification using a visual-linguistic multi-modal model.

Deep learning technology has been successfully applied to classification tasks, providing new solutions for waste classification. Many algorithms for waste classification have been proposed by researchers. For example, Majchrowska et al. [6] proposed a two-stage detector for waste localization and classification using semi-supervised training, achieving good results. White et al. [7] designed WasteNet, deployable on edge devices, to assist smart trash bins in making quick decisions. Sheng et al. [8] implemented an Internet of Things (IoT) system for trash bins based on the LoRa communication protocol, enabling intelligent waste management systems capable of detection and classification. Furthermore, to improve the efficiency and accuracy of waste classification, researchers proposed a transfer learning-based waste image classification algorithm [9], and created the NWNUTRASH dataset to address issues such as a single background and small sample size in existing waste datasets. Another study [10] introduced GCDN-Net, a comprehensive solution for single-label and multi-label classification tasks, achieving outstanding performance on the GIGO dataset. Additionally, [11] presented the MMTrash dataset for mixed solid waste and integrated the CBAM and CotNet networks to develop the real-time high-precision DSYOLO-Trash algorithm. However, for kitchen waste, challenges such as significant background interference, varying sizes, and diverse categories have limited research to object detection in kitchen waste processing pipelines [12]. To address these challenges, this paper employs the self-constructed MLKW dataset for the experimental study of multi-label classification of kitchen waste.

The removal of impurities from kitchen waste is a critical step in downstream waste processing. It involves separating out foreign materials such as plastic bags, takeout containers, and beverage cartons, which is essential to ensure the purity of

Manuscript received Month xx, 2xxx; revised Month xx, xxxx; accepted Month x, xxxx. This work was supported in part by the Project of the National Key Research and Development Program of China (2021YFC1910402), Hunan Provincial Natural Science Foundation (2026JJ90097), and Scientific Research Fund of Hunan Provincial Education Department (25A0672). (*Corresponding author: Qiaokang Liang*).

Hai Qin is with the School of Electronic Information, Hunan First Normal University, Changsha, 410205, China and National Engineering Research Center of Robot Vision Perception and Control, School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China (email: qinhai@hnu.edu.cn).

Qiaokang Liang, Jintao Li, Wei Sun and Yaonan Wang are with the National Engineering Research Center of Robot Vision Perception and Control, School of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China (email: qiaokang@hnu.edu.cn; lj1825@hnu.edu.cn; wei_sun@hnu.edu.cn; yaonan@hnu.edu.cn).

Dan Zhang is with the Department of Mechanical Engineering The Hong Kong Polytechnic University Kowloon, Hong Kong (email: dan.zhang@polyu.edu.hk).

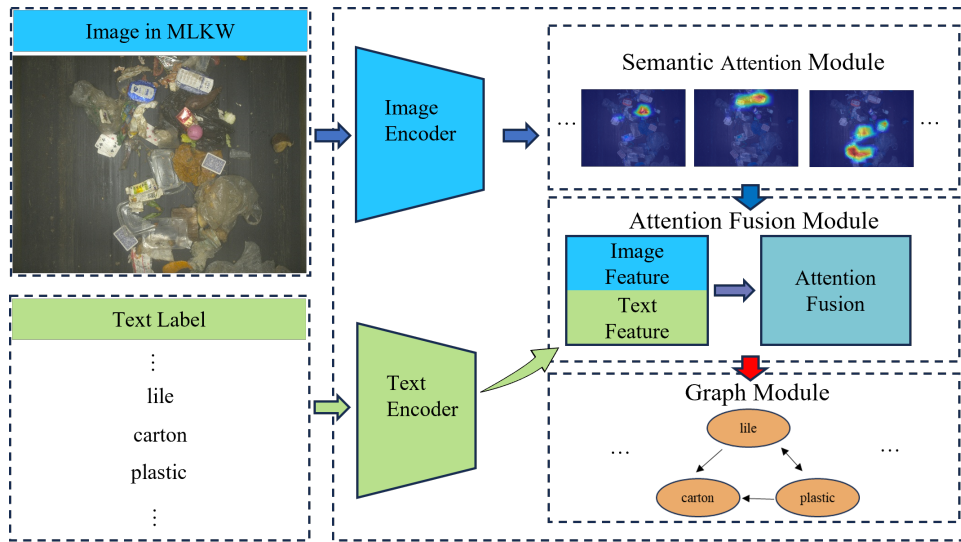


Fig. 1: Overall framework of the MM-SAG model. The model takes images from the MLKW dataset and their corresponding textual labels as inputs, which are processed by an image encoder and a text encoder to extract visual features and label semantic embeddings, respectively. A semantic attention module is then introduced in the visual branch to highlight key visual regions that are highly correlated with label semantics, producing semantic-aware visual feature representations. Subsequently, visual and textual features are aligned and interact across modalities within the attention fusion module, yielding label-semantic-guided visual representations. The fused features are further fed into a graph module, where label co-occurrence relationships are modeled to construct a label correlation graph, thereby enhancing the modeling of label dependencies for the final multi-label classification task.

organic waste. Traditionally, this task is performed manually; however, due to harsh working conditions, unpleasant odors, and potential health hazards, manual sorting is often inefficient and poses risks to human workers. With advancements in technology, machine-based sorting has emerged as a promising alternative. Compared to manual operations, mechanical systems offer higher efficiency and lower operational costs. Typically, these systems rely on visual cameras—serving as the “eyes” of the machine—to capture image data and apply image recognition algorithms to distinguish impurity categories, thereby ensuring the stability of the subsequent sorting process. Considering that kitchen waste may simultaneously contain multiple types of impurities, multi-label classification provides an effective solution for improving machine recognition performance. Unlike traditional single-label classification, multi-label image classification—a key task in computer vision—aims to accurately predict all relevant categories present in an image. By leveraging multi-label classification, sorting systems can achieve more comprehensive and accurate recognition, ultimately assisting human operators in completing their tasks more efficiently.

Multi-label image classification [13], [14], [15] was first introduced in the 1990s and has since been extensively explored by researchers over the past three decades. Multi-label classification refers to the task where each sample may be associated with multiple labels, in contrast to traditional single-label classification where each sample corresponds to only one label. In real-world visual classification tasks, images often contain multiple semantic entities, making multi-label classification highly relevant. However, compared to single-label classifi-

cation, multi-label classification presents additional challenges due to the dependencies or mutual exclusiveness among labels. It also requires addressing issues such as class imbalance and semantic overlap. As one of the fundamental tasks in computer vision, multi-label image classification aims to determine which categories are present in a given image. An inherent subtask of this process is to discover and exploit the relationships among labels. In this context, the Semantic-Aware Module is designed to mine the instances present in an image. It first performs an initial classification to obtain the categories of instances contained in the sample. Then, in the subsequent processing, it focuses on the features of the identified categories. Therefore, it is termed a semantic-aware module [16], [17]. Multi-label image classification, as a fundamental task in computer vision, primarily focuses on identifying the categories present in an image. A key subtask is uncovering the relationships between these labels. There are currently two main approaches to discovering these potential label correlations. The first approach, exemplified by [13], constructs potential co-occurrence relationships between labels using graph models. The second approach, seen in [14], [15], derives instance-level co-occurrence relationships by extracting semantic features from each image. The advancement of multi-label classification tasks is closely linked to the deep research on graph convolutional networks (GCN). For instance, [13] proposed a GCN-based multi-label recognition algorithm, ML-GCN, which leverages graph convolution to uncover co-occurrence relationships among labels, achieving excellent results. However, the co-occurrence relationships generated based on labels are global and can be too coarse for

individual images. In contrast, the ADD-GCN algorithm proposed by [15] introduces dynamic graph convolution, creating a unique label graph relationship for each image, but it lacks consideration for multi-modality. To address this, the M3TR algorithm [18] constructs a triplet learning relationship within and between modalities, mainly through interactions between visual contexts and between visual and language modalities. However, this algorithm overlooks the potential relationship mining capabilities of graph convolution. This paper leverages current research hotspots in deep learning, specifically visual-linguistic models, to explore multi-label classification on the MLKW dataset using a visual-linguistic model approach.

Based on the visual-linguistic model CLIP [19] architecture, this paper explores and addresses two key issues: (1) Mining potential relationships between labels, and (2) Fusion methods for visual and linguistic modalities. We propose the Multi-Modal Semantic-Aware Graph framework, as shown in Fig. 1. This framework introduces a semantic-aware module for instance-level label relationship mining. The captured semantic features are then used to generate a label correlation matrix through graph convolution, improving the efficiency and effectiveness of label correlation mining. For the fusion of visual and linguistic modalities, we design an improved multi-head attention mechanism module. This module realigns and re-encodes visual and textual features, further enhancing feature extraction capabilities. Our proposed method achieves excellent results on the MLKW dataset and demonstrates generalization capabilities on the public datasets VOC2007 and MS-COCO.

Overall, the contributions of this paper are threefold:

- 1) A novel multi-label classification model framework, MM-SAG, designed based on the CLIP architecture. It integrates the text modality with the visual modality for feature fusion and combines two different label mining schemes: instance-level mining through a semantic-aware module and potential relationship construction through graph convolution.
- 2) An improved multi-head attention mechanism that uses a set of learnable parameters as queries to focus locally on beneficial information in feature vectors, increasing the amount of effective information in feature vectors. A residual structure is designed to prevent information loss, enhancing the fusion of visual and linguistic modalities and further improving feature extraction, especially for textual feature embeddings.
- 3) A semantic-aware module to guide the mining of label correlations through a semantic feature map, enabling instance-level label relationship mining. Experiments show that our method achieves excellent results on the multi-label classification dataset MLKW.

II. RELATED WORKS

In recent years, with the rapid development of deep neural networks and the emergence of large-scale datasets such as MS-COCO [20], PASCAL VOC [21], and ImageNet [22], image recognition technology has experienced significant advancements. A notable shift has been observed from traditional

single-label classification to more challenging and technically demanding multi-label classification. The primary task of multi-label image classification is to predict multiple object labels present in an image. Benefiting from the remarkable success of convolutional neural networks (CNN) in single-label classification tasks [23], [24], [25], numerous CNN-based approaches have been applied to multi-object recognition and have achieved promising results.

A. Multi-Label Image Classification

The task of multi-label image classification has attracted significant attention from researchers in deep learning and computer vision. Currently, the mainstream approaches focus on capturing and modeling the dependencies between various labels within an input image to effectively identify all target labels. Early works primarily utilized convolutional neural networks to establish label relationship graphs [26], [27]. For example, Wang et al. [26] proposed an RNN- CNN framework to predict final scores and derive label dependencies accordingly. Some studies have sought to uncover these relationships by introducing Transformer models [28] and long short-term memory networks (LSTM) [29], leveraging attention mechanisms to focus on specific regions. Xiao et al. [30] effectively improved both the efficiency and reconstruction quality of remote sensing image super-resolution by selectively retaining key tokens. However, these methods often explore relationships sequentially within localized regions, occasionally overlooking global information. Recent advancements have addressed these limitations. Zhao et al. [31] proposed a dual relational graph framework based on Transformers to mitigate the inherent deficiency of CNNs and RNNs in utilizing global information. This framework employs semantic-aware constraints and structural positional constraints in a complementary manner for label recognition. Weng et al. [32] directly calculated global label correlations based on label co-occurrence frequencies. Wu et al. [33] aligned visual models with pre-trained textual models to extract global visual information. Graph convolutional networks, which can integrate spatial information to address the underutilization of label positional relationships, have also been increasingly adopted in multi label classification tasks in recent years.

B. Graph Convolutional Networks

Graph convolutional networks function similarly to feature extractors, leveraging node or edge features to perform tasks such as node classification and graph classification on graph-structured data. Compared to standard CNNs, GCNs are better suited for capturing global information when the input object is an image. In recent years, using GCNs to establish global label relationship graphs has become a popular approach among researchers [13], [15], [34], [35], [36]. Chen et al. [13] utilized GCNs to generate more accurate semantic annotations and incorporated them into word embeddings, thereby obtaining improved label relationship graphs. Ye et al. [15] extended the GCN framework by introducing a dynamic graph convolutional network (DGCN) model to construct individual label dependencies for each image. Chen et al. [34]

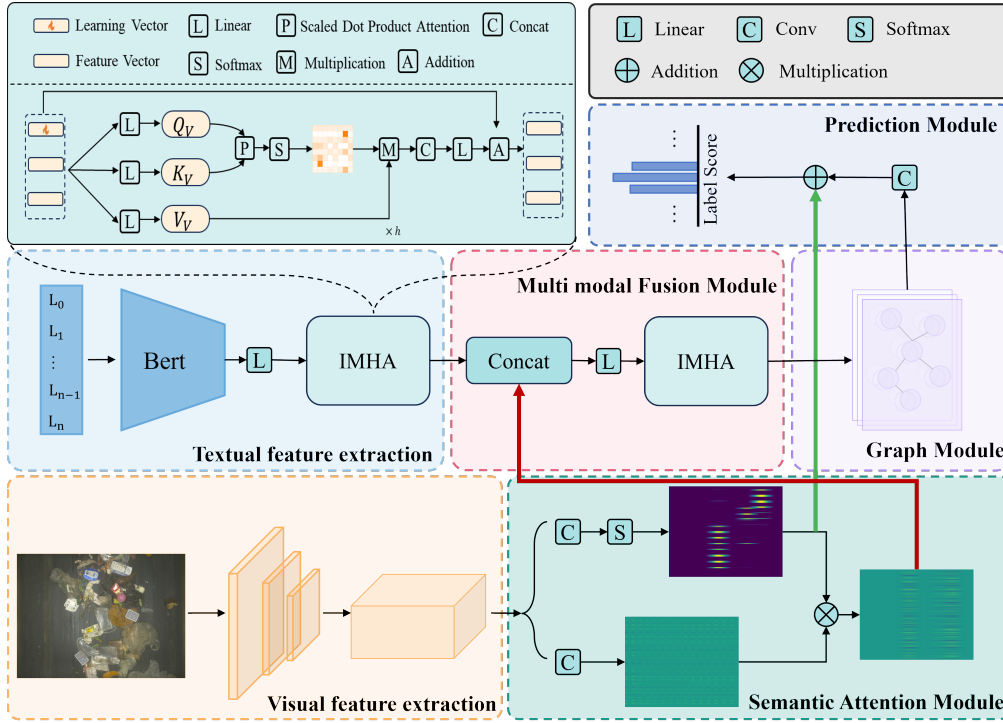


Fig. 2: Detailed architecture and computational flow of the MM-SAG framework. The input image and its corresponding label texts are fed into a visual backbone network and a text encoder, respectively, to extract a visual feature map and label semantic embeddings. In the visual branch, a Semantic Attention Module is introduced, where the intermediate outputs are visualized as heatmaps, indicating the degree of model attention to key image regions under different semantic guidance, i.e., the label-semantic-weighted responses over the visual features. These processes progressively produce semantically enhanced visual representations. Subsequently, the label semantic embeddings enhanced by the improved multi-head attention mechanism interact and align with the semantic-aware visual features in the attention fusion module, yielding label-specific, semantically aligned visual features. The fused features are then fed into a graph module, where label co-occurrence relationships are modeled to construct a label correlation graph and generate label-discriminative vectors for multi-label classification. Finally, the classifier predicts the possible labels present in the input image based on these discriminative representations.

designed the predictive graph convolutional network (PGCN) model to construct directed graphs of "classes" and learn the interdependencies between labels at the class level. Liang et al. [37] applied a dynamic graph convolutional module to adaptively capture semantic-aware regions, further enhancing recognition capabilities. Leveraging the advantage of GCNs in capturing global information, this study employs GCNs to generate label correlation matrices, thereby improving the efficiency and effectiveness of label dependency mining.

C. Visual Modality and Linguistic Modality

In recent years, integrating information from images (visual modality) and text (linguistic modality) has emerged as a research hotspot due to its enormous potential in multimodal understanding tasks [38], [39], [40]. Gao et al. [38] proposed a knowledge-enhanced visual and linguistic model to improve multimodal fake news detection. Wu et al. [41] introduced a multimodal prototype regularization method for few-shot fine-tuning in CLIP-based image classification, effectively leveraging both image and text characteristics. Wang et al. [42] presented a regularization-constrained multimodal fusion method for image captioning, enabling better integration of

inter- and intra-modal information. For image classification tasks, Fan et al. [43] proposed a model comprising multi-modal attention and hierarchical fusion for intangible cultural heritage image classification. Kong et al. [44] introduced a joint classification method based on spectral prompt tuning for hyperspectral images and LiDAR data. However, few studies have employed multimodal fusion methods for multi-label image classification tasks. Motivated by this gap, this study explores an image-text modality fusion approach to address the multi-label classification of kitchen waste images with complex, unstructured backgrounds.

III. METHODOLOGY

The proposed MM-SAG framework operates as follows: First, images and the corresponding textual labels from the dataset are used as inputs. These inputs are mapped to label semantic embeddings and visual feature maps through a backbone network and a language model, respectively. A semantic-aware module is introduced after the visual feature map to obtain a visual-semantic feature map. The label semantic embeddings are further enriched with semantic information using an improved multi-head attention mechanism, serving as

inputs to the subsequent stage. Second, the improved multi-head attention mechanism is employed to achieve fusion between the two modalities, associating relevant visual features with each label's semantic embedding and enhancing the visual representation. Next, the fused outputs are fed into a graph module to construct the label co-occurrence matrix and generate corresponding label-discriminative vectors for multi-label classification. Finally, a trained classifier predicts the potential labels present in the input image. Fig.2 illustrates the detailed workflow of the proposed MM-SAG framework.

A. Semantic Attention Module

The input dataset contains labels denoted as $L = \{L_1, L_2, L_3, \dots, L_N\}$, where N is the number of categories. By concatenating these labels, a label sentence is formed as the input to the language model, generating label embeddings denoted as t , where $t \in \mathbb{R}^{d_t}$ and d_t represents the dimensionality of the label embeddings. The obtained label embeddings are passed through a fully connected layer and subsequently forwarded to an improved multihead attention module for further enhancement. The resulting enhanced label embeddings are denoted as:

$$T = IMHA(Linear(t)), \quad (1)$$

The label embedding t is processed through a fully connected layer *Linear* and an improved multi-head attention module IMHA, which will be detailed in the next section. Equation (1) represents the final step in the label information processing pipeline, producing an enhanced label representation T that incorporates deep semantic understanding and internal relational modeling. This enriched representation provides a crucial foundation for effective interaction with the visual features X in subsequent stages. For the input image I , its image features are extracted using a CNN backbone network and denoted as X , where $X \in \mathbb{R}^{H \times W \times D}$, with H , W , and D representing the height, width, and number of channels of the image features, respectively. The goal of the Semantic Attention Module (SAM) is to capture the category-specific feature representations contained within the visual image. As shown in Fig.2, the features X extracted by the visual backbone serve as the input to the SAM. In the upper section of the SAM, the *Convblock* acts as a classifier and employs *Softmax*($\cdot$) for normalization. This classifier is designed to generate the category feature map $P = [p_1, p_2, p_3, \dots, p_N]$, where N represents the number of categories and $P \in \mathbb{R}^{(H \times W \times N)}$. Correspondingly, in the lower section of the SAM, a *Convblock* transforms $X \in \mathbb{R}^{(H \times W \times D)}$ into $X' \in \mathbb{R}^{(H \times W \times D')}$, where X' and D' are the transformed features and the corresponding channel dimension, respectively. Finally, a weighted aggregation of P and X' is performed to obtain the category-specific feature representations $V = [v_1, v_2, v_3, \dots, v_N] \in \mathbb{R}^{(D' \times N)}$. As illustrated in Fig. 2, each element v_N in the category-specific feature representation V can be expressed as the weighted sum of X' , as defined in Equation2. This approach ensures that the

resulting category-specific feature v_N purposefully aggregates features associated with category n .

$$v_N = P_N^T \cdot X' = \sum_{i=1}^H \sum_{j=1}^W P_{(i,j)}^n \cdot X'_{(i,j)} \quad (2)$$

here, $P_{(i,j)}^n$ and $X'_{(i,j)}$ represent the category feature map representation and the feature vector of the feature map at position (i, j) , respectively. It achieves the transformation from generic image features X to class-specific discriminative features V through category-specific attention weights v_N . The feature v_N serves as a critical bridge connecting visual representations and semantic labels, and is ultimately used for classification. The visualization of the attention map P_N^T can intuitively reveal which regions of the image the model considers most important for recognizing the corresponding category.

B. Improved Multi-Head Attention Module

This section provides a detailed explanation of the Improved Multi-Head Attention (IMHA) module, which primarily serves to enhance feature information and facilitate multimodal feature fusion.

The structure of the improved multi-head attention module is illustrated in the dashed box in the upper left corner of Fig. 2. Given three input vectors, as depicted on the far left of the figure, the vectors are denoted from top to bottom as *query*(Q), *key*(K) and *value*(V). Here, Q is defined as a set of learnable parameters initialized using a standard normal distribution, where each element is randomly sampled from a Gaussian distribution with a mean of 0 and a standard deviation of 1, while K and V are derived from the feature vectors. The computation of IMHA is similar to the multi-head attention mechanism in Transformer [45].

Initially, Q , K , and V are separately projected using distinct linear projections W_i^Q , W_i^K and W_i^V , mapping them to subspaces of dimensions d_q , d_k and d_v , respectively, for each attention head i (where $i \in [1, h]$). This process yields Q_V , K_V , and V_V . Subsequently, scaled dot-product attention is performed in parallel on Q_V , K_V , and V_V .

Specifically, the scaled dot-product attention computes the dot product between the scaled query and key vectors. Subsequently, the softmax function is applied to obtain the corresponding attention weights A . The mathematical expression for this attention mechanism is as follows:

$$A = \text{Softmax} \left(\frac{Q_V K_V^T}{\sqrt{d_k}} \right) \quad (3)$$

$$\text{Attention}(Q_V, K_V, V_V) = A V_V \quad (4)$$

here, Q_V , K_V , and V_V are the matrices obtained after the linear projections of Q , K , and V , respectively. $\sqrt{d_k}$ serves as the scaling factor, where d_k is the projection dimension mentioned earlier.

The multi-head attention mechanism module consists of multiple subspaces *head_i* (where $i \in [1, h]$), and the corresponding expression is as follows:

$$\text{Multihead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W \quad (5)$$

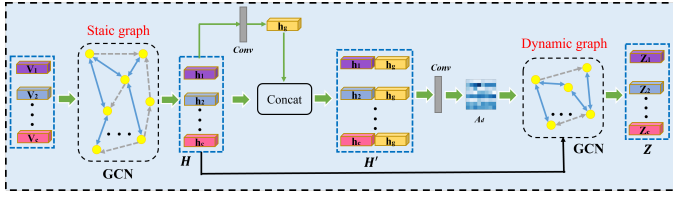


Fig. 3: Detailed structure and computational workflow of the GCN module.

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (6)$$

where $W_i^Q \in \mathbb{R}^{(d_{model} \times d_k)}$, $W_i^K \in \mathbb{R}^{(d_{model} \times d_k)}$, $W_i^V \in \mathbb{R}^{(d_{model} \times d_v)}$, and $W \in \mathbb{R}^{(hd_v \times d_{model})}$. Here, d_{model} denotes the dimension of the input feature vector for the module, d_k and d_v are the projection dimensions mentioned earlier, and W_i^Q, W_i^K, W_i^V and W represent the projection matrices. Since we use a set of learnable parameters as the *query*(Q), the objective is to focus locally on the beneficial information in the feature vector, thereby enhancing the effective information content within the feature vector. To this end, we designed a residual structure to prevent information loss and incorporated Dropout and layer normalization operations to mitigate overfitting during the training process, as expressed in the following equation:

$$ImAttention = Q \oplus LN(Dropout(Linear(Multihead(Q, K, V)))) \quad (7)$$

here, $\oplus$ denotes element-wise addition, while LN and $Dropout$ represent Layer Normalization and Dropout operations, respectively. The core of Equation 7 lies in the improved attention module, which leverages a set of learnable query parameters Q to guide the model in focusing on key information within the input. After computing multi-head attention, the module applies Dropout to prevent overfitting and LN to stabilize training. Additionally, it innovatively introduces a residual connection by directly adding the original query information back to the output. This design effectively mitigates the loss of valuable original information that may occur during the attention process, thereby enhancing the richness and robustness of the resulting feature representations.

C. GCN Module

This section introduces the Graph Module, whose primary task is to uncover the potential contributing labels in an image. Among the numerous methods related to Graph Modules, this study improves upon the DGCN proposed in [15]. This module addresses the insufficient attention to individual instances observed in the GCN presented in [13], focusing more on the co occurrence labels at the individual level. The detailed structure of this module is illustrated in Fig.3.

Based on the module fusion process in Fig. 2, the DGCN takes the fused feature vector $V = [v_1, v_2, \dots, v_C]$ as input and propagates it sequentially through the static GCN (SGCN)

and dynamic GCN (DGCN). The relevant computations for the SGCN are shown as follows:

$$H = LReLU(A_s V W_s) \quad (8)$$

where H represents the feature set after SGCN computation, $H = [h_1, h_2, \dots, h_C] \in \mathbb{R}^{(C \times D_1)}$, where C denotes the number of channels, indicating that the input features share the same dimensionality. The feature information contained in a single element of H is represented here by D_1 , which serves as an intermediate representation for transition purposes only. $LReLU(\bullet)$ denotes the activation function *LeakyReLU*, while A_s and W_s refer to the randomly initialized associated matrix and state update matrix during training, respectively. Then, H is passed through convolution and concatenation operations to obtain $H' \in \mathbb{R}^{(2D_1 \times C)}$, where C has the same meaning as above. D_2 is formed by concatenating two D_1 features from H , resulting in the relationship $D_2 = 2D_1$. Next, dynamic graph convolution is performed on H . Unlike static graph convolution, the associated matrix A_d in DGCN is dynamically constructed based on the input features, which helps enhance the model's expressive power. Here, $Z \in \mathbb{R}^{(C \times D_2)}$, and the computation is as follows:

$$Z = f(\delta(W_A H') H W_d) \quad (9)$$

where $f(\cdot)$ denotes the *LeakyReLU* activation function, $\delta(\cdot)$ represents the *Sigmoid* activation function, $W_d \in \mathbb{R}^{D_1 \times D_2}$ has the same meaning as W_s , both referring to the state update parameters, $W_A \in \mathbb{R}^{C \times 2D_1}$ represents the transformation parameters of A_d , and $H' \in \mathbb{R}^{2D_1 \times C}$ as previously described. Additionally, H' can be expressed as:

$$H' = [(h_1; h_g), (h_2; h_g), \dots, (h_c; h_g)] \quad (10)$$

Where $h_g \in \mathbb{R}^{D_1}$ is obtained by applying global pooling followed by a convolutional layer to H . The dynamic adjacency matrix A_d here is generated on an individual basis, making it unique. This is a key feature that enables the DGCN to perform local individual modeling.

D. Prediction Module and Loss Function

The final part of the MM-SAG algorithm framework is the Prediction Module, whose main task is to perform classification prediction for the given image. As mentioned earlier, after the GCN, the category representation is $Z = [z_1, z_2, \dots, z_c] \in \mathbb{R}^{C \times D_2}$. Since each vector in Z contains rich category feature information, it is passed through a simple binary classifier to predict the scores. It is important to note that, to enrich the image's category features, the category feature map $P = [p_1, p_2, p_3, \dots, p_N]$ from the *SAM* module is introduced, and global spatial pooling is applied to obtain another prediction score. By aggregating and averaging these two prediction scores, a more reliable result is obtained.

During the training process, a common loss function for multi-label classification is used to supervise the final prediction scores. The formula for the loss function is as follows:

$$\mathcal{L}(y, s) = \sum_{c=1}^C y^c \log(\sigma(s^c)) + (1 - y^c) \log(1 - \sigma(s^c)) \quad (11)$$

where $\sigma(\cdot)$ is the *Sigmoid*($\cdot$) function. The purpose of this formula is to supervise the model's multi-label classification predictions. For each category c in the dataset, the formula independently computes the predicted score s^c , applies the Sigmoid function to obtain the probability $\sigma(s^c)$, and then measures the loss based on the difference between $\sigma(s^c)$ and the ground-truth label y^c (0 or 1). This loss function directly drives the model to learn and optimize the parameters of the GCN module, the SAM module, and the classifier, enabling the predicted probability for the presence of each category to closely approximate the ground-truth labels. In doing so, it facilitates accurate multi-label image classification.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets and Evaluation Metrics

This paper focuses on multi-label image classification of kitchen waste. We selected a subset of images taken from a sorting conveyor at an urban kitchen waste processing plant to create the kitchen waste multi-label classification dataset, MLKW. The original images were captured with a resolution of 3256×2724 . To ensure the representativeness of kitchen waste across different seasons, the MLKW dataset was captured in batches from different time periods over the course of a year. The final selection includes 3771 images from the spring, 4735 from the summer, 2130 from the autumn, and 4358 from the winter. Due to the presence of many duplicate or irrelevant images in the original dataset, a secondary filtering process was applied, resulting in 3107 images for dataset creation. The MLKW dataset contains eight categories: 'waste bag', 'metal', 'shoe', 'plastic', 'bottle', 'carton', 'lile', and 'glass'. The dataset is split into training and testing sets in an 8:2 ratio, with the images resized to 640×480 . The statistical analysis of the MLKW dataset is presented in Fig. 4. As shown in subfigure (a), the top-left chart illustrates the sample distribution across eight kitchen waste categories, revealing a clear class imbalance. The top-right co-occurrence matrix highlights the frequency with which different labels appear together in the same image, with darker shades indicating stronger co-occurrence. The bottom-left heatmap visualizes the spatial distribution of object instances, showing that most objects are located near the center of the image. The bottom-right plot depicts the distribution of object widths and heights, indicating that the majority of targets are relatively small in size, with dimensions concentrated in the lower range. Subfigure (b) presents a label co-occurrence graph, where each node represents a waste category, and edges indicate the frequency of co-occurrence between categories. The number and thickness of the edges reflect the strength of semantic relationships—more and thicker edges denote higher co-occurrence and stronger label correlation. Finally, subfigure (c) provides an overview of the relative proportions of each category in the dataset. This analysis helps to better understand the data characteristics and informs the design of our multi-label classification model.

To evaluate the generalization ability of the proposed method, experiments were also conducted on the PASCAL VOC 2007 multi-label dataset and the MSCOCO dataset.

PASCAL VOC 2007 is a popular benchmark for evaluating the performance of multi-label image classification models. It includes 20 category labels and 9963 images, divided into a training set with 5011 images and a test set with the remaining 4952 images used for model evaluation. MS COCO is another popular benchmark for validating the performance of multi-label image classification models. It is larger and more comprehensive than PASCAL VOC 2007, containing 80 category labels and 122,581 images. The training set includes 82,081 images, the validation set contains 40,504 images, with an average of 2.9 category labels per image.

To ensure fairness in the experimental results comparison, we use the evaluation metrics specified in papers [18], [13], [15], including Average Precision (AP), Mean Average Precision (mAP), Overall Precision (OP), Overall Recall (OR), Overall F1 Score (OF1), Category Precision (CP), Category Recall (CR), and Category F1 Score (CF1). These evaluation metrics are calculated as follows:

$$OP = \frac{\sum_i N_i^t}{\sum_i N_i^p} \quad (12)$$

$$CP = \frac{1}{c} \sum_i \frac{N_i^t}{N_i^p} \quad (13)$$

$$OR = \frac{\sum_i N_i^t}{\sum_i N_i^g} \quad (14)$$

$$CR = \frac{1}{c} \sum_i \frac{N_i^t}{N_i^g} \quad (15)$$

$$OF1 = \frac{2 \times OP \times OR}{OP + OR} \quad (16)$$

$$CF1 = \frac{2 \times CP \times CR}{CP + CR} \quad (17)$$

where N_i^t represents the number of images correctly predicted for the i -th label, N_i^p represents the number of images predicted for the i -th label, and N_i^g represents the number of true images for the i -th label.

B. Experimental Details

The MM-SAG algorithm proposed in this paper is based on an improved visual-language model. The label text is vectorized using a pre-trained BERT model to serve as the feature embedding for the labels. It is important to note that for the same dataset, the input label text for the language model in this work is fixed, and the resulting label embeddings are also unique. For a fair comparison, the backbone network used in this work is the ResNet101 pre trained model on ImageNet. Additionally, during the training process, the input image sizes are adjusted to a fixed dimension, and the DGCN module employs the nonlinear activation function LeakyReLU with a slope of 0.2. Some hyperparameters used during the training process are as follows: the optimizer is SGD with a momentum of 0.9 and a weight decay coefficient of 0.0001. The batch size is set to 16, the initial learning rate for the MHA and DGCN modules is 0.5, and the initial learning rate for the backbone network is 0.05. The model is trained for 50 epochs, with

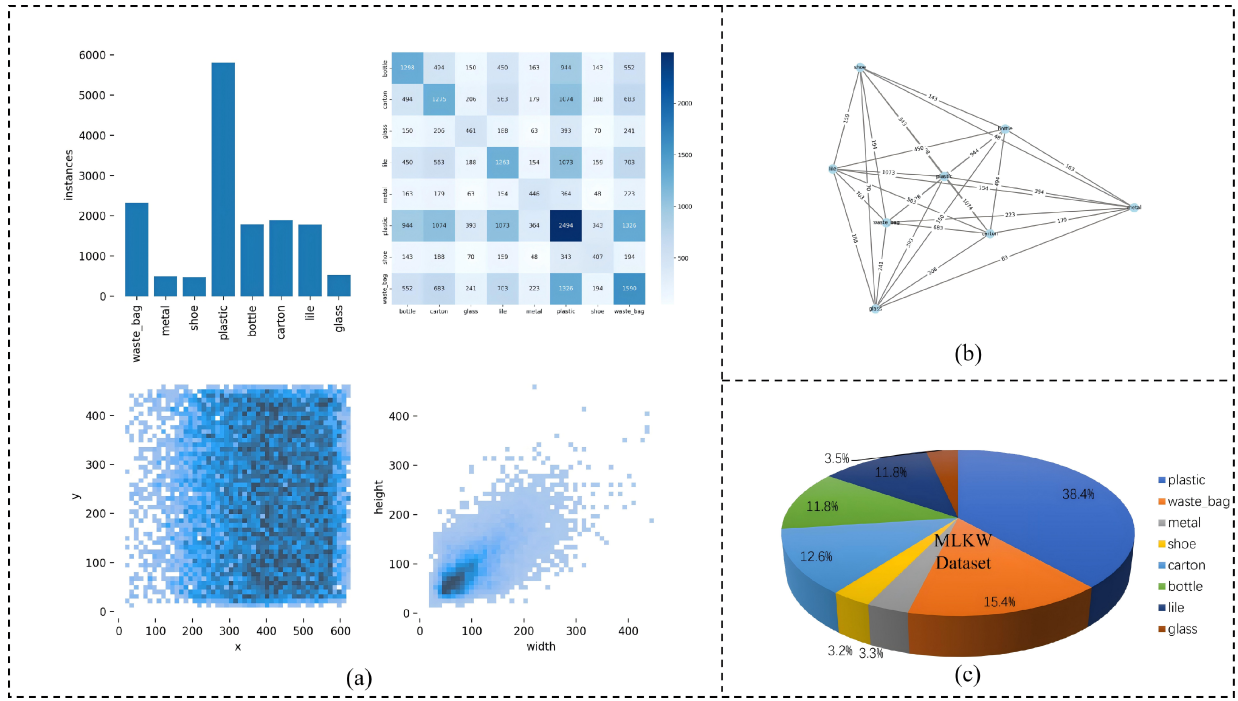


Fig. 4: Statistical analysis of the MLKW dataset. (a) Displays the sample distribution across eight categories of kitchen waste, category co-occurrence matrix, object spatial location heatmap, and object size distribution; (b) Label co-occurrence graph, where nodes represent labels and edges (with weights) indicate co-occurrence frequency, revealing semantic relationships; (c) Overall category proportion distribution across the dataset.

learning rate decay applied at the 30th and 40th epochs with a decay factor of 0.1. All experiments are conducted on an NVIDIA RTX 3090 and implemented using the PyTorch 1.13 framework [46].

C. Comparison Experiments and Analysis

To validate the effectiveness and generalization capability of the proposed framework, the experiments in this study are divided into two parts, each designed with different objectives and handled accordingly. In the first part, we conduct comprehensive comparisons on the MLKW dataset against several state-of-the-art multi-label classification models. The methods compared are categorized into unimodal and multimodal models based on the format of their input. The unimodal models include ResNet101 [47], ADD-GCN [15], and TDRG [31], while the multimodal models include ML-GCN [13], SGTN [48], SSGRL [14], M3TR [18], MSRN [49], MGTN [50], Query2Lab [51], and CLIP [19]. All these methods, including ML-GCN, are reproduced under identical experimental settings, including backbone architecture, input format, and optimization parameters, to ensure fair and valid comparisons. To further verify the transferability and robustness of our proposed framework, the second part involves generalization experiments on two widely used public datasets: MS-COCO and VOC2007. In this part, the results of the baseline methods are directly cited from their original publications, and we strictly follow the same experimental protocols and settings as described in those works when implementing our proposed method, thereby ensuring a fair evaluation of generalization performance.

1) *Experiments on the MLKW Dataset:* The MLKW dataset is built from real-world kitchen waste sorting scenarios, with 8 categories and a total of 15,098 labels, averaging 4.8 labels per image. Additionally, the dataset contains images with complex compositions, significant variations between samples, and intricate backgrounds, making high-precision classification a challenging task. Table I shows the multi label classification results on the MLKW dataset using different advanced algorithms, with mAP as the core metric and other metrics as supplementary indicators. In the comparison methods, the proposed method outperforms others by a noticeable margin. Our MM-SAG model, when compared with the same feature extraction backbone ResNet101, enhances label correlation exploration by utilizing semantic feature maps, which improves the model's feature extraction capability, especially in terms of text feature embeddings.

Analysis of Table I reveals that graph convolution-based methods, such as ADD-GCN, SSGRL, and ML-GCN, generally perform well on MLKW. This is mainly because graph convolution, as a supplement to text or other modalities, helps to uncover possible co-occurring labels in images. The experimental results indicate that the MM-SAG framework achieves significant performance improvements, with a 1.3% increase in the key metric, mAP, and significant advantages in CP, CR, CF1, OP, OR, and OF1. These results demonstrate the superiority of the multi-modal semantic attention graph MM-SAG learning framework.

2) *Generalization Performance Validation Experiment:* In this section, we conduct a comparison based on the MM-SAG framework using the public datasets Pascal VOC 2007 and

TABLE I: Experimental Results of MM-SAG and Other Methods on the MLKW Dataset.

Modality	Methods	(Size)	All						Top3						mAP
			CP	CR	CF1	OP	OR	OF1	CP	CR	CF1	OP	OR	OF1	
Single Modal	ResNet-101[47]	(448,448)	0.725	0.553	0.627	0.724	0.616	0.664	0.741	0.527	0.616	0.740	0.586	0.654	0.721
	ADD-GCN[15]	(448,448)	0.760	0.631	0.690	0.806	0.732	0.767	0.803	0.566	0.664	0.841	0.672	0.757	0.751
	TDRG[31]	(448,448)	0.586	0.461	0.516	0.822	0.667	0.736	0.578	0.426	0.490	0.832	0.627	0.715	0.650
Multi Modal	ML-GCN[13]	(448,448)	0.754	0.609	0.674	0.740	0.683	0.710	0.776	0.583	0.666	0.761	0.654	0.703	0.738
	SGTN[48]	(448,448)	0.699	0.526	0.600	0.808	0.665	0.730	0.740	0.473	0.578	0.823	0.615	0.704	0.682
	SSGRL[14]	(448,448)	0.792	0.688	0.737	0.806	0.744	0.774	0.821	0.611	0.700	0.831	0.673	0.745	0.818
	M3TR[18]	(448,448)	0.446	0.365	0.401	0.753	0.558	0.641	0.456	0.337	0.388	0.772	0.526	0.626	0.584
	MSRN[49]	(448,448)	0.760	0.680	0.718	0.790	0.759	0.774	0.804	0.601	0.688	0.822	0.686	0.748	0.786
	MGTN[50]	(448,448)	0.757	0.580	0.657	0.804	0.703	0.750	0.799	0.533	0.639	0.829	0.657	0.733	0.735
	Query2Lab[51]	(448,448)	-	-	-	-	-	-	-	-	-	-	-	-	0.422
	Clip[19]	(448,448)	-	-	-	-	-	-	-	-	-	-	-	-	0.515
	Ours	(448,448)	0.813	0.714	0.760	0.829	0.773	0.800	0.848	0.633	0.725	0.857	0.700	0.770	0.831

TABLE II: Experimental results of MM-SAG and other methods on the VOC2007 dataset.

Methods	aero	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	motor	person	plant	sheep	sofa	train	tv	mAP
CNN-RNN[26]	96.7	83.1	94.2	92.8	61.2	82.1	89.1	94.2	64.2	83.6	70.0	92.4	91.7	84.2	93.7	59.8	93.2	75.3	99.7	78.6	84.0
RLSD[52]	96.4	92.7	93.8	94.1	71.2	92.5	94.2	95.7	74.3	90.0	74.2	95.4	96.2	92.1	97.9	66.9	93.5	73.7	97.5	87.6	88.5
VeryDeep[53]	98.9	95.0	96.8	95.4	69.7	90.4	93.5	96.0	74.2	86.6	87.8	96.0	96.3	93.1	97.2	70.0	92.1	80.3	98.1	87.0	89.7
ResNet-101[47]	99.5	97.7	97.8	96.4	65.7	91.8	96.1	97.6	74.2	80.9	85.0	98.4	96.5	95.9	98.4	70.1	88.3	80.2	98.9	89.2	89.9
FeV+LV[54]	97.9	97.0	96.6	94.6	73.6	93.9	96.5	95.5	73.7	90.3	82.8	95.4	97.7	95.9	98.6	77.6	88.7	78.0	98.3	89.0	90.6
HCP[55]	98.6	97.1	98.0	95.6	75.3	94.7	95.8	97.3	73.1	90.2	80.0	97.3	96.1	94.9	96.3	78.3	94.7	76.2	97.9	91.5	90.9
RNN-Attention[27]	98.6	97.4	96.3	96.2	75.2	92.4	96.5	97.1	76.5	92.0	87.7	96.8	97.5	93.8	98.5	81.6	93.7	82.8	98.6	89.3	91.9
Atten-Reinforce[56]	98.6	97.1	97.1	95.5	75.6	92.8	96.8	97.3	78.3	92.2	87.6	96.9	96.5	93.6	98.5	81.6	93.1	83.2	98.5	89.3	92.0
VGG(Re-weighted)[13]	99.4	97.4	98.0	97.0	77.9	92.4	96.8	97.8	80.8	93.4	87.2	98.0	97.3	95.8	98.8	79.4	95.3	82.2	99.1	91.4	92.8
ML-GCN(Re-weighted)[13]	99.5	98.5	98.6	98.1	80.8	94.6	97.2	98.2	82.3	95.7	86.4	98.2	98.4	96.7	99.0	84.7	96.7	84.3	98.9	93.7	94.0
Ours	99.6	98.7	98.0	98.5	80.6	95.2	97.7	99.0	79.7	97.5	88.5	98.6	99.1	99.1	99.0	83.2	98.9	79.8	99.6	94.7	94.3

MSCOCO to verify the generalization capability of our framework. However, it is important to note that since the framework proposed in this paper is designed based on the complex background of the urban kitchen waste dataset (MLKW), when compared to the latest multi-label classification methods from Section 4.3.1, there may still be some gap compared to the latest methods on the public datasets Pascal VOC 2007 and MSCOCO. Therefore, in this section, we focus on transferability experiments to validate the generalization performance of the proposed algorithm.

The experimental results on Pascal VOC 2007 are shown in Table II. The proposed method demonstrates certain advantages in terms of AP values across multiple categories, particularly in 14 categories, including cow, motor, and horse. By using the MM-SAG framework, the classification performance is further improved, with mAP increasing by 4.4% compared to the ResNet network. The overall performance of the proposed method is 0.3% better than ML GCN, which demonstrates the generalization and scalability of the proposed approach.

The experimental results on MS COCO are shown in

Table III. Compared to Pascal VOC 2007, MS COCO has a larger scale, more categories, and is more challenging. The proposed method shows an improvement of 1.2% in mAP over ML-GCN. It also outperforms other methods in evaluation metrics such as CR, CF1, OR, and OF1, indicating superior performance on multi-label classification tasks in complex backgrounds. This also proves the robustness of the method on other types of datasets.

D. Ablation Study

This section conducts ablation experiments on the proposed framework using the MLKW dataset.

1) *Selection of Experimental Parameters:* In terms of hyperparameter settings, we followed conventional practices for initialization and fine-tuning. We found that the learning rate has a significant impact on both the training stability and the final performance of the model. Therefore, we conducted a learning rate sensitivity ablation study to determine an appropriate value. The results are shown in Fig. 5.

According to the experimental results, when the initial learning rate of the backbone network is set to 0.05, our

TABLE III: Experimental results of MM-SAG and other methods on the MSCOCO dataset.

Methods	All							Top-3						
	mAP	CP	CR	CF1	OP	OR	OF1	CP	CR	CF1	OP	OR	OF1	
CNN-RNN[26]	61.2	-	-	-	-	-	-	66.0	55.6	60.4	69.2	66.4	67.8	
RNN-Attention[27]	-	-	-	-	-	-	-	79.1	58.7	67.4	84.0	63.0	72.0	
Order-Free RNN[57]	-	-	-	-	-	-	-	71.6	54.8	62.1	74.2	62.2	67.7	
ML-ZSL[58]	-	-	-	-	-	-	-	74.1	64.5	69.0	-	-	-	
SRN[47]	77.1	81.6	65.4	71.2	82.7	69.9	75.8	85.2	58.8	67.4	87.4	62.5	72.9	
ResNet-101[47]	77.3	80.2	66.7	72.8	83.9	70.8	76.8	84.1	59.4	69.7	89.1	62.8	73.6	
Multi-Evidence[59]	-	80.4	70.2	74.9	85.2	72.5	78.4	84.5	62.2	70.6	89.1	64.3	74.7	
ML-GCN(Binary)[13]	80.3	81.1	70.1	75.2	83.8	74.2	78.7	84.9	61.3	71.2	88.8	65.2	75.2	
ML-GCN(Re-weighted)[13]	83.0	85.1	72.0	78.0	85.8	75.4	80.3	89.2	64.1	74.6	90.5	66.5	76.7	
Ours	84.2	82.1	76.0	78.9	83.8	78.8	81.2	87.1	66.0	75.1	89.3	68.0	77.2	

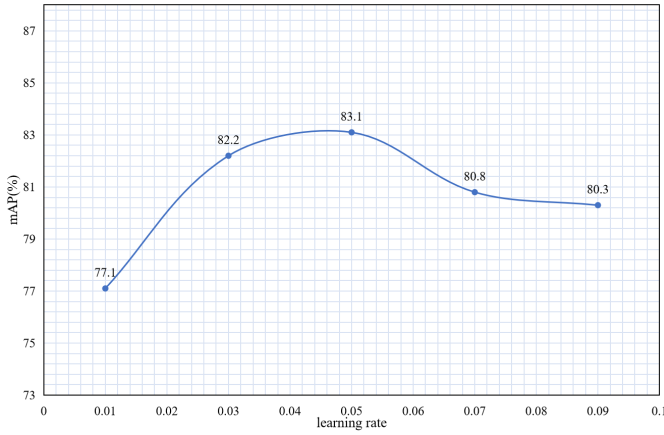


Fig. 5: Sensitivity analysis of learning rate.

proposed IMHA model achieves the highest mAP of 83.1% on the MLKW dataset. This result indicates that a learning rate of 0.05 strikes a good balance between training speed and convergence stability in our model. On one hand, a relatively large learning rate helps the model escape local minima and accelerates convergence; on the other hand, it is not so high as to cause training oscillations or convergence failure. As a result, it effectively enhances the model's feature learning capability and overall classification performance.

2) *Evaluating the Effectiveness of Key Modules*: The first step is to validate the effectiveness of each module in the proposed framework. The experimental results are shown in Table IV. It is important to note that all experiments in the table are based on a multi-modal framework with ResNet101 as the visual backbone. To assess the impact of each module on overall performance, we conducted a series of ablation experiments. From the experimental results in Table IV, we observe that when the Fusion module is omitted, the mAP of SAM+Graph is reduced by 1.1% compared to SAM+Fusion, and SAM+Graph performs 2.3% worse in comparison. This highlights the critical role of the Fusion module in multi-modal feature fusion. Furthermore, analysis of the SAM module in the visual modality revealed its decisive impact on algorithm performance. When SAM, Fusion, and Graph modules are considered independently, the performance is best when only

the SAM module is used, with mAP increased by 28.4% and 3.4% compared to Fusion and Graph modules, respectively. Experiments with the Graph, Fusion+Graph, and SAM+Graph configurations demonstrated that the Graph module contributes to further improving the framework's performance, with the best results achieved when all three modules—SAM, Fusion, and Graph—are combined.

In addition, we analyze each module from the perspective of model complexity and computational efficiency, including the number of parameters, FLOPs, inference time, and peak GPU memory consumption, as reported in the last four columns of Table IV. It can be observed that the introduction of the SAM and Fusion modules leads to moderate increases in computational cost and memory usage due to the additional attention operations and multimodal feature interactions; however, the overall overhead remains well controlled.

Specifically, while achieving the best performance, the complete MM-SAG framework contains 617.4MB parameters and requires 1058.86G FLOPs, with an inference time of 157.13 ms per image and a peak GPU memory consumption of 1317.19MB, which are comparable to those of several simplified variants. In contrast, removing the Graph module slightly reduces the computational and memory costs but results in a noticeable degradation in mAP, indicating the critical role of graph-based label relationship modeling in performance improvement. Models incorporating only the Graph or Fusion module exhibit lower computational overhead, yet their classification performance is significantly limited, suggesting that performance gains do not merely stem from increased model complexity. Furthermore, a comparison between the Fusion+Graph and SAM+Graph configurations shows that the SAM module achieves a favorable balance between performance improvement and computational cost. Overall, the proposed MM-SAG framework delivers substantial performance gains with only a limited additional computational burden, demonstrating an effective trade-off between efficiency and accuracy.

3) *Evaluation of the Effectiveness of the IMHA Module Construction*: This paper proposes an improved multi-head attention mechanism. To verify the effectiveness of this structure, an ablation experiment was conducted on the IMHA module using the MLKW dataset. As shown in Table V, setting the query parameters as learnable increased the mAP from 82.8%

TABLE IV: Ablation Study of Each Module in the MM-SAG Framework.

SAM	Fusion	Graph	mAP	CP	CR	CF1	OP	OR	OF1	Params	FLOPs	Inference time	GPU peak memory
✓			81.5	80.0	69.1	74.1	82.5	76.0	79.1	617.4 MB	1058.86 G	154.84 ms	1320.69 MB
✓	✓		82.6	80.1	72.1	75.9	81.8	77.7	79.7	629.48 MB	1059.39 G	158.11 ms	1332.79 MB
✓		✓	80.3	79.0	68.3	73.3	81.7	75.2	78.3	627.48 MB	1059.01 G	157.00 ms	1327.03 MB
		✓	82.1	79.6	69.5	74.2	81.6	75.9	78.6	629.48 MB	1059.66 G	156.53 ms	1329.79 MB
		✓	53.1	44.4	51.5	47.7	64.6	73.1	68.6	617.4 MB	1059.32 G	157.03 ms	1317.7 MB
		✓	78.1	72.9	71.1	72.0	76.3	76.6	76.4	629.48 MB	555.76 G	87.43 ms	1304.04 MB
✓	✓	✓	83.1	81.3	71.4	76.0	82.9	77.3	80.0	617.4 MB	1058.86 G	157.13 ms	1317.19 MB

TABLE V: Ablation Study of Certain Structures in the IMHA Module.

Module	Query		Residual	mAP
	Vector	learnable		
IMHA	✓			82.8%
		✓		83.0%
		✓	✓	83.1%

to 83.0%. Furthermore, with the introduction of the residual structure, the performance improved further. These significant performance improvements validate the effectiveness of these two structural designs.

4) *Evaluation of the Combined Effect of Dynamic and Static Graph Convolutions.*: In this work, we adopt a hybrid graph modeling strategy that combines SGCN and DGCN to jointly capture global label correlations and image-specific adaptive relationships. To verify the effectiveness of this design, ablation experiments were conducted on the MLKW dataset by comparing three settings: using only SGCN, using only DGCN, and integrating both graph structures. The quantitative results are reported in table VI.

TABLE VI: Ablation results of GCN module combinations.

Methods	mAP
SGCN	81.31
DGCN	80.62
SGCN+DGCN	83.1

As shown in table VI, the model achieves an mAP of 81.31% when only SGCN is employed and 80.62% when only DGCN is used. In contrast, the combined static–dynamic graph design significantly improves the performance to 83.1%, yielding a maximum gain of approximately 2.5 percentage points over the single-graph configurations. This result clearly demonstrates the complementary nature of static and dynamic graph modeling in capturing label dependencies. From the perspective of graph structure modeling, the SGCN constructs a global label correlation graph based on dataset-level co-occurrence statistics, resulting in a stable and consistent graph topology. Such a structure effectively captures long-term and common semantic dependencies between frequently co-occurring categories (e.g., plastic–bottle and carton–waste bag), thereby providing reliable global priors for label relation-

ship reasoning. However, due to its fixed structure, the static graph lacks the flexibility to adapt to image-specific semantic variations. In contrast, the DGCN dynamically generates label relationships conditioned on the visual–semantic features of each input image, leading to graph structures that vary across samples. Visualization of intermediate dynamic graphs shows that this mechanism can adaptively emphasize task-relevant label connections in complex scenes or under heavy occlusion, enhancing local contextual modeling. Nevertheless, relying solely on dynamic graphs may introduce noise, especially in cases with limited or ambiguous visual evidence, resulting in less stable label dependency modeling. By jointly integrating static and dynamic graph structures, the proposed model benefits from both global semantic consistency and image-level adaptive relationship modeling. Under the constraint of global co-occurrence priors, the dynamic graph refines label dependencies in a sample-specific manner, leading to more discriminative feature representations. Feature visualization results further indicate that the collaborative graph modeling strategy yields more compact intra-class feature distributions and clearer inter-class boundaries, particularly for confusing or long-tail categories such as glass and carton.

Overall, both quantitative results and qualitative analyses of intermediate graph structures and feature representations confirm that the synergistic combination of SGCN and DGCN significantly enhances label dependency modeling and classification performance, validating the effectiveness and necessity of the proposed graph-based design for multi-label classification tasks.

5) *Comparison between the IMHA module and the standard multi-head attention mechanism.*: We conducted an ablation study to compare the proposed Improved Multi-Head Attention (IMHA) module with the standard multi-head attention mechanism (denoted as SrcMHA). As shown in Table VII, on the MLKW dataset, the mAP achieved by SrcMHA is 70.30%, whereas our IMHA module reaches 83.1%, demonstrating a significant performance improvement.

TABLE VII: Comparison results between IMHA and the standard multi-head attention mechanism.

Methods	mAP
SrcMHA	70.30
OurIMHA	83.1

This result clearly validates the effectiveness of our improvements. The IMHA module enhances semantic awareness

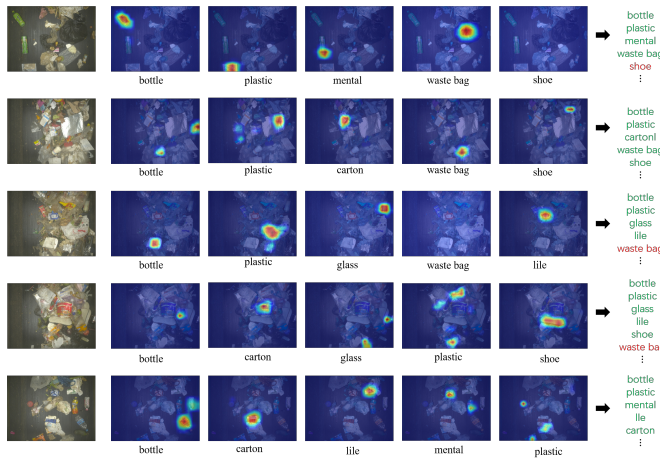


Fig. 6: Heatmap visualization on the MLKW dataset, where the green font on the right indicates presence and red indicates absence.

and cross-modal feature alignment by introducing a learnable query vector Q and a semantics-guided alignment mechanism. These enhancements enable the model to better capture label-specific representations and significantly improve multi-label classification performance.

E. Visualization Analysis

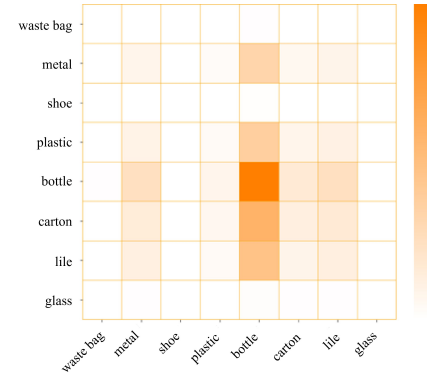
In this section, we visualize the semantic feature perception maps in *SAM* and the weight values (A) in the IMHA module, to illustrate how the semantic feature maps locate semantic targets and how multimodal fusion focuses on specific category information.

We visualize the original images and their corresponding category labels through heatmaps to demonstrate how the SAM module locates semantic feature information in the image. Fig.6 shows some examples, where each row represents the original image, its corresponding category feature heatmap, and the predicted labels. Upon observing the examples, it is evident that the heatmaps accurately locate the regions corresponding to the categories present in the image. For categories not present in the image, the heatmap response for the corresponding labels is relatively low. This validates that the SAM module has sufficient discriminative ability to accurately identify the semantic features contained in the image.

In addition, the entire framework utilizes the IMHA module for the multimodal fusion process. The figure demonstrates the visualization of the attention to different categories during feature fusion. For the input image in Fig.7, its true labels are metal, plastic, bottle, carton, and tile. On the right is the visualization of the corresponding weight values A . It can be observed that the values of $A^{mental,bottle}$, $A^{plastic,bottle}$, $A^{bottle,bottle}$, $A^{carton,bottle}$, and $A^{tile,bottle}$ are ranked relatively high, especially $A^{bottle,bottle}$, which further indicates that the feature fusion through IMHA helps focus better on the key category information.



(a) input image



(b) the A matrix in IMHA

Fig. 7: Visualization of the weight matrix A in the IMHA module.

V. CONCLUSION

This paper addresses the challenging problem of urban kitchen waste classification in complex backgrounds by designing a semantic-aware module for instance-level label relationship mining. Combined with an improved multi-head attention mechanism, we propose a multi-modal semantic-aware graph (MM-SAG) algorithm for high-precision multi-label classification of kitchen waste images. This method captures semantic features and generates a label-related matrix through graph convolution, significantly enhancing the efficiency and effectiveness of label correlation mining. Additionally, by realigning and recombining visual and textual features, the feature extraction capability is further improved. Experimental results demonstrate that the proposed method achieves superior performance on the multi-label classification dataset MLKW, as well as on two public datasets, VOC2007 and MS-COCO.

Despite the notable performance gains, the incorporation of graph-based modeling and multi-module collaboration inevitably increases computational complexity, and the recognition performance on long-tail categories still leaves room for improvement. Future work will focus on developing parameter-efficient and lightweight graph neural network designs, as well as incorporating strategies to address class imbalance, in order to further improve efficiency, robustness, and recognition accuracy for minority classes.

REFERENCES

- [1] R. Sindhu, E. Gnansounou, S. Rebello, P. Binod, S. Varjani, I. S. Thakur, R. B. Nair, and A. Pandey, "Conversion of food and kitchen waste to value-added products," *Journal of environmental management*, vol. 241, pp. 619–630, 2019.
- [2] S. Das, S.-H. Lee, P. Kumar, K.-H. Kim, S. S. Lee, and S. S. Bhattacharya, "Solid waste management: Scope and the challenge of sustainability," *Journal of cleaner production*, vol. 228, pp. 658–678, 2019.
- [3] X. Liu, J. Shen, Y. Guo, S. Wang, B. Chen, L. Luo, and H. Zhang, "Technical progress and perspective on the thermochemical conversion of kitchen waste and relevant applications: A comprehensive review," *Fuel*, vol. 331, p. 125803, 2023.
- [4] T. R. Kelley and P. M. Walker, "Bacterial concentration reduction of food waste amended animal feed using a single-screw dry-extrusion process," *Bioresour technology*, vol. 67, no. 3, pp. 247–253, 1999.
- [5] J. Ma, T. H. Duong, M. Smits, W. Verstraete, and M. Carballa, "Enhanced biomethanation of kitchen waste by different pre-treatments," *Bioresour technology*, vol. 102, no. 2, pp. 592–599, 2011.
- [6] S. Majchrowska, A. Mikołajczyk, M. Ferlin, Z. Klawikowska, M. A. Plantykowski, A. Kwasigroch, and K. Majek, "Deep learning-based waste detection in natural and urban environments," *Waste Management*, vol. 138, pp. 274–284, 2022.
- [7] G. White, C. Cabrera, A. Palade, F. Li, and S. Clarke, "Wastenet: Waste classification at the edge for smart bins," *arXiv preprint arXiv:2006.05873*, 2020.
- [8] T. J. Sheng, M. S. Islam, N. Misran, M. H. Baharuddin, H. Arshad, M. R. Islam, M. E. Chowdhury, H. Rmili, and M. T. Islam, "An internet of things based smart waste management system using lora and tensorflow deep learning model," *IEEE Access*, vol. 8, pp. 148 793–148 811, 2020.
- [9] Q. Zhang, Q. Yang, X. Zhang, Q. Bao, J. Su, and X. Liu, "Waste image classification based on transfer learning and convolutional neural network," *Waste Management*, vol. 135, pp. 150–157, 2021.
- [10] M. M. Hossen, A. Ashraf, M. Hasan, M. E. Majid, M. Nashbat, S. B. A. Kashem, A. K. A. Kunju, A. Khandakar, S. Mahmud, and M. E. Chowdhury, "Gcdn-net: Garbage classifier deep neural network for recyclable urban waste management," *Waste Management*, vol. 174, pp. 439–450, 2024.
- [11] W. Ma, H. Chen, W. Zhang, H. Huang, J. Wu, X. Peng, and Q. Sun, "Dsyolo-trash: An attention mechanism-integrated and object tracking algorithm for solid waste detection," *Waste Management*, vol. 178, pp. 46–56, 2024.
- [12] H. Qin, L. Shu, L. Zhou, S. Deng, H. Xiao, W. Sun, Q. Liang, D. Zhang, and Y. Wang, "Active learning-detr: Cost-effective object detection for kitchen waste," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–15, 2024.
- [13] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Multi-label image recognition with graph convolutional networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 5177–5186.
- [14] T. Chen, M. Xu, X. Hui, H. Wu, and L. Lin, "Learning semantic-specific graph representation for multi-label image recognition," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 522–531.
- [15] J. Ye, J. He, X. Peng, W. Wu, and Y. Qiao, "Attention-driven dynamic graph convolutional network for multi-label image recognition," in *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXI 16*. Springer, 2020, pp. 649–665.
- [16] J. Ma, G.-S. Xie, F. Zhao, and Z. Li, "Afanet: Adaptive frequency-aware network for weakly-supervised few-shot semantic segmentation," *IEEE Transactions on Multimedia*, pp. 1–11, 2025.
- [17] Y. Tang, Y. Ma, and T. Zhang, "Semantic-aware network for natural language tracking," *IEEE Transactions on Circuits and Systems for Video Technology*, pp. 1–11, 2025.
- [18] J. Zhao, Y. Zhao, and J. Li, "M3tr: Multi-modal multi-label recognition with transformer," in *Proceedings of the 29th ACM international conference on multimedia*, 2021, pp. 469–477.
- [19] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PMLR, 2021, pp. 8748–8763.
- [20] Y. Li, C. Huang, C. C. Loy, and X. Tang, "Human attribute recognition by deep hierarchical contexts," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part VI 14*. Springer, 2016, pp. 684–700.
- [21] M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman, "The pascal visual object classes (voc) challenge," *International journal of computer vision*, vol. 88, pp. 303–338, 2010.
- [22] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 2009, pp. 248–255.
- [23] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 580–587.
- [24] R. Girshick, "Fast r-cnn," *arXiv preprint arXiv:1504.08083*, 2015.
- [25] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *IEEE transactions on pattern analysis and machine intelligence*, vol. 39, no. 6, pp. 1137–1149, 2016.
- [26] J. Wang, Y. Yang, J. Mao, Z. Huang, C. Huang, and W. Xu, "Cnn-rnn: A unified framework for multi-label image classification," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2285–2294.
- [27] Z. Wang, T. Chen, G. Li, R. Xu, and L. Lin, "Multi-label image recognition by recurrently discovering attentional regions," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 464–472.
- [28] M. Jaderberg, K. Simonyan, A. Zisserman *et al.*, "Spatial transformer networks," *Advances in neural information processing systems*, vol. 28, 2015.
- [29] A. Graves and A. Graves, "Long short-term memory," *Supervised sequence labelling with recurrent neural networks*, pp. 37–45, 2012.
- [30] Y. Xiao, Q. Yuan, K. Jiang, J. He, C.-W. Lin, and L. Zhang, "Ttst: A top-k token selective transformer for remote sensing image super-resolution," *IEEE Transactions on Image Processing*, vol. 33, pp. 738–752, 2024.
- [31] J. Zhao, K. Yan, Y. Zhao, X. Guo, F. Huang, and J. Li, "Transformer-based dual relation graph for multi-label image recognition," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 163–172.
- [32] W. Weng, B. Wei, W. Ke, Y. Fan, J. Wang, and Y. Li, "Learning label-specific features with global and local label correlation for multi-label classification," *Applied Intelligence*, vol. 53, no. 3, pp. 3017–3033, 2023.
- [33] X. Wu, Q.-Y. Jiang, Y. Yang, Y.-F. Wu, Q.-G. Chen, and J. Lu, "Tai++: Text as image for multi-label image classification by co-learning transferable prompt," *arXiv preprint arXiv:2405.06926*, 2024.
- [34] Z.-M. Chen, X.-S. Wei, P. Wang, and Y. Guo, "Learning graph convolutional networks for multi-label recognition and applications," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 6, pp. 6969–6983, 2021.
- [35] K. Li and W. Ye, "Semi-supervised node classification via graph learning convolutional neural network," *Applied Intelligence*, vol. 52, no. 11, pp. 12 724–12 736, 2022.
- [36] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 7, pp. 5966–5978, 2020.
- [37] Q. Liang, J. Li, H. Qin, M. Liu, X. Xiao, D. Zhang, Y. Wang, and D. Zhang, "Mha-dgcn: multi-head attention-driven dynamic graph convolutional lightweight network for multi-label image classification of kitchen waste," *Applied Intelligence*, vol. 54, no. 24, pp. 13 057–13 074, 2024.
- [38] X. Gao, X. Wang, Z. Chen, W. Zhou, and S. C. Hoi, "Knowledge enhanced vision and language model for multi-modal fake news detection," *IEEE Transactions on Multimedia*, 2024.
- [39] B. Zhang, Z. Yuan, H. Xu, and K. Gao, "Crossmodal translation based meta weight adaption for robust image-text sentiment analysis," *IEEE Transactions on Multimedia*, 2024.
- [40] Y. Liu, G. Li, and L. Lin, "Cross-modal causal relational reasoning for event-level visual question answering," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 10, pp. 11 624–11 641, 2023.
- [41] Q. Wu, J. Qi, D. Zhang, H. Zhang, and J. Tang, "Fine-tuning for few-shot image classification by multimodal prototype regularization," *IEEE Transactions on Multimedia*, 2024.
- [42] L. Wang, H. Chen, Y. Liu, and Y. Lyu, "Regular constrained multimodal fusion for image captioning," *IEEE Transactions on Circuits and Systems for Video Technology*, 2024.
- [43] T. Fan, H. Wang, and S. Deng, "Intangible cultural heritage image classification with multimodal attention and hierarchical fusion," *Expert Systems with Applications*, vol. 231, p. 120555, 2023.

- [44] Y. Kong, Y. Cheng, Y. Chen, and X. Wang, "Joint classification of hyperspectral image and lidar data based on spectral prompt tuning," *IEEE Transactions on Geoscience and Remote Sensing*, 2024.
- [45] A. Vaswani, "Attention is all you need," *Advances in Neural Information Processing Systems*, 2017.
- [46] A. Paszke, S. Gross, S. Chintala, G. Chanan, E. Yang, Z. DeVito, Z. Lin, A. Desmaison, L. Antiga, and A. Lerer, "Automatic differentiation in pytorch," 2017. [Online]. Available: <https://api.semanticscholar.org/CorpusID:40027675>
- [47] F. Zhu, H. Li, W. Ouyang, N. Yu, and X. Wang, "Learning spatial regularization with image-level supervisions for multi-label image classification," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 5513–5522.
- [48] X.-S. Vu, D.-T. Le, C. Edlund, L. Jiang, and H. D. Nguyen, "Privacy-preserving visual content tagging using graph transformer networks," in *Proceedings of the 28th ACM International Conference on Multimedia*, 2020, pp. 2299–2307.
- [49] J. Li, F. Fang, K. Mei, and G. Zhang, "Multi-scale residual network for image super-resolution," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 517–532.
- [50] H. D. Nguyen, X.-S. Vu, and D.-T. Le, "Modular graph transformer networks for multi-label image classification," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 35, no. 10, 2021, pp. 9092–9100.
- [51] S. Liu, L. Zhang, X. Yang, H. Su, and J. Zhu, "Query2label: A simple transformer way to multi-label classification," *arXiv preprint arXiv:2107.10834*, 2021.
- [52] J. Zhang, Q. Wu, C. Shen, J. Zhang, and J. Lu, "Multilabel image classification with regional latent semantic dependencies," *IEEE Transactions on Multimedia*, vol. 20, no. 10, pp. 2801–2813, 2018.
- [53] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.
- [54] H. Yang, J. Tianyi Zhou, Y. Zhang, B.-B. Gao, J. Wu, and J. Cai, "Exploit bounding box annotations for multi-label object recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 280–288.
- [55] Y. Wei, W. Xia, M. Lin, J. Huang, B. Ni, J. Dong, Y. Zhao, and S. Yan, "Hcp: A flexible cnn framework for multi-label image classification," *IEEE transactions on pattern analysis and machine intelligence*, vol. 38, no. 9, pp. 1901–1907, 2015.
- [56] T. Chen, Z. Wang, G. Li, and L. Lin, "Recurrent attentional reinforcement learning for multi-label image recognition," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 32, no. 1, 2018.
- [57] S.-F. Chen, Y.-C. Chen, C.-K. Yeh, and Y.-C. Wang, "Order-free rnn with visual attention for multi-label classification," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 32, no. 1, 2018.
- [58] C.-W. Lee, W. Fang, C.-K. Yeh, and Y.-C. F. Wang, "Multi-label zero-shot learning with structured knowledge graphs," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 1576–1585.
- [59] W. Ge, S. Yang, and Y. Yu, "Multi-evidence filtering and fusion for multi-label classification, object detection and semantic segmentation based on weakly supervised learning," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 1277–1286.